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Marx et al.

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(54) **SYSTEM FOR DIRECTLY MEASURING THE DEPTH OF A HIGH ASPECT RATIO ETCHED FEATURE ON A WAFER**

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(22) Filed: **Apr. 29, 2016**

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G01B 11/22	(2006.01)
G01B 11/06	(2006.01)

(52) **U.S. Cl.**

CPC ***G01B 11/22*** (2013.01); ***G01B 11/0633*** (2013.01); ***G01B 11/0675*** (2013.01); *G01B 2210/56* (2013.01)

(58) **Field of Classification Search**

CPC H01L 21/00; H01L 22/12; H01L 21/66; H01J 37/3255; H01J 21/3065; G01B 9/02; G01B 11/22

See application file for complete search history.

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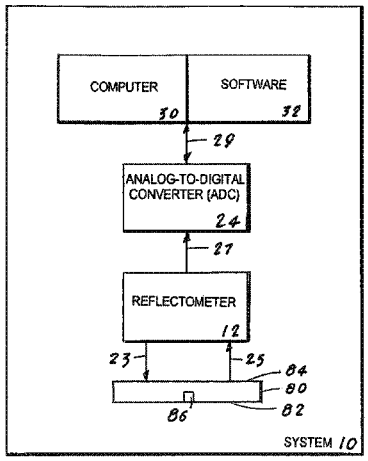
(57) **ABSTRACT**

A system (**10**) for directly measuring the depth of a high aspect ratio etched feature on a wafer (**80**) that includes an etched surface (**82**) and a non-etched surface (**84**). The system (**10**) utilizes an infrared reflectometer (**12**) that in a preferred embodiment includes a swept laser (**14**), a fiber circulator (**16**), a photodetector (**22**) and a combination collimator (**18**) and an objective lens (**20**). From the objective lens (**20**) a focused incident light (**23**) is produced that is applied to the non-etched surface (**84**) of the wafer (**80**). From the wafer (**80**) is produced a reflected light (**25**) that is processed through the reflectometer (**12**) and applied to an ADC (**24**) where a corresponding digital data signal (**29**) is produced. The digital data signal (**29**) is applied to a computer (**30**) that, in combination with software (**32**), measures the depth of the etched feature that is then viewed on a display (**34**).

15 Claims, 6 Drawing Sheets



Related U.S. Application Data

continuation of application No. 13/134,716, filed on Jun. 15, 2011, now Pat. No. 8,649,016, which is a continuation-in-part of application No. 12/456,781, filed on Jun. 23, 2009, now abandoned.

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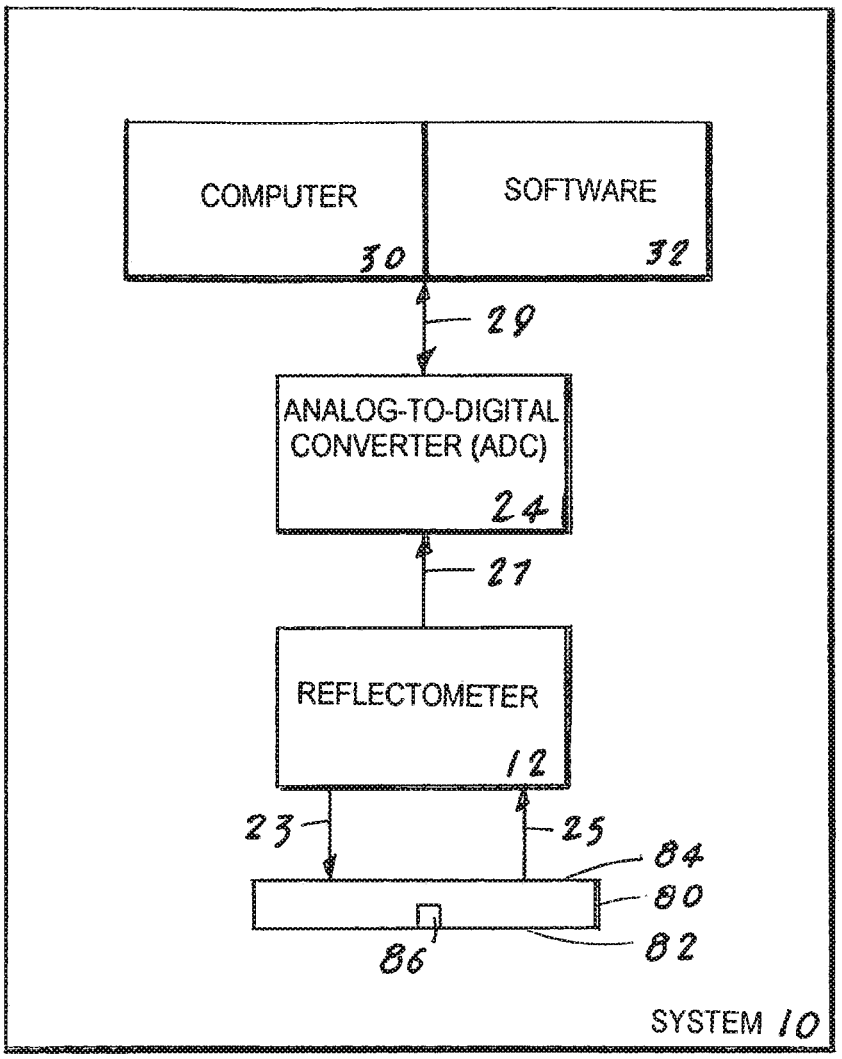
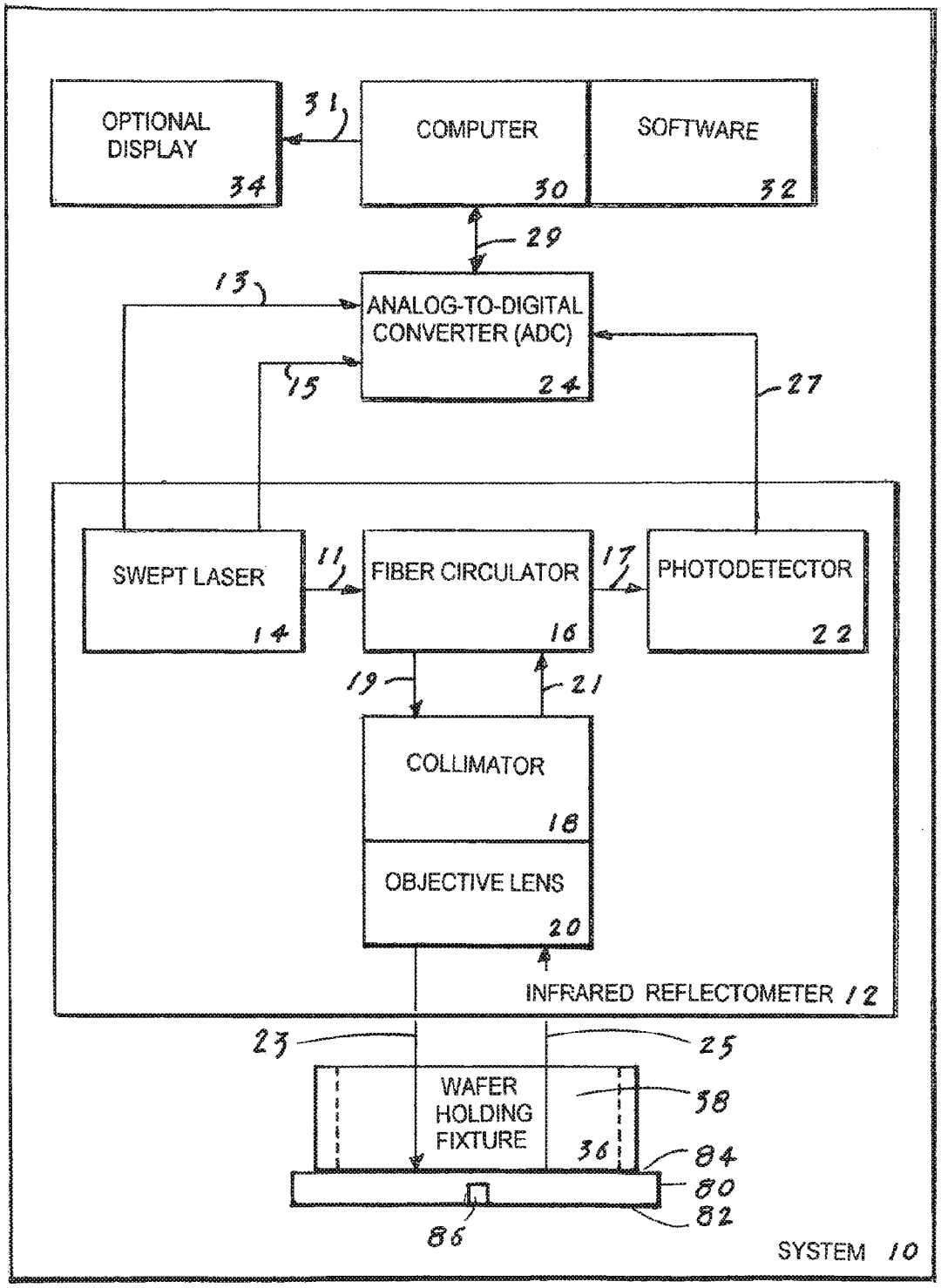

COMPUTER
30
SOFTWARE
32
29
ANALOG-TO-DIGITAL CONVERTER (ADC)
24
27
REFLECTOMETER
12
23
25
84
80
86
82
SYSTEM 10

Fig. 1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FIBER CIRCULATOR
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COLLIMATOR
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OBJECTIVE LENS
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WAFER HOLDING FIXTURE
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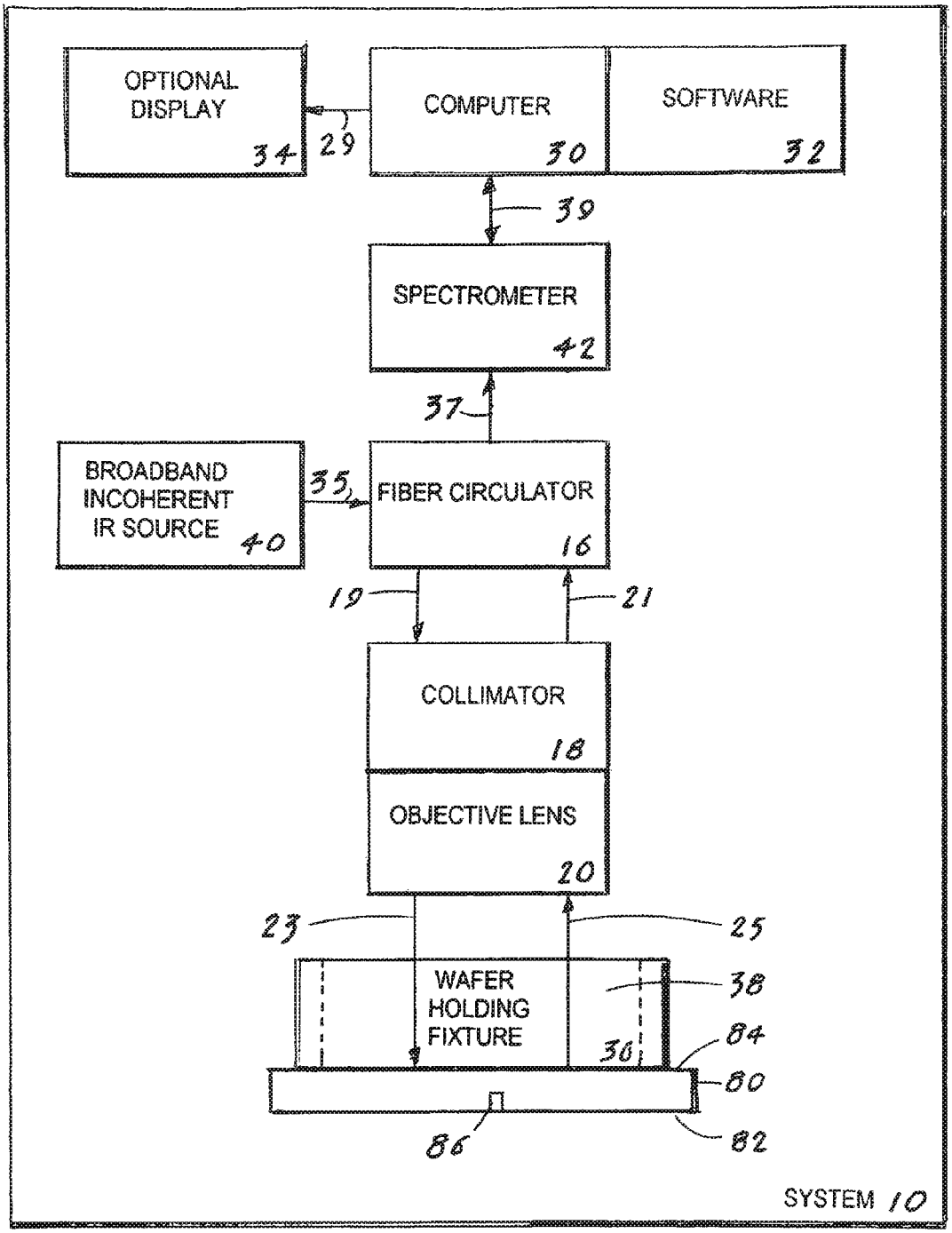
SYSTEM 10

Fig. 2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COLLIMATOR
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OBJECTIVE LENS
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WAFER HOLDING FIXTURE
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36
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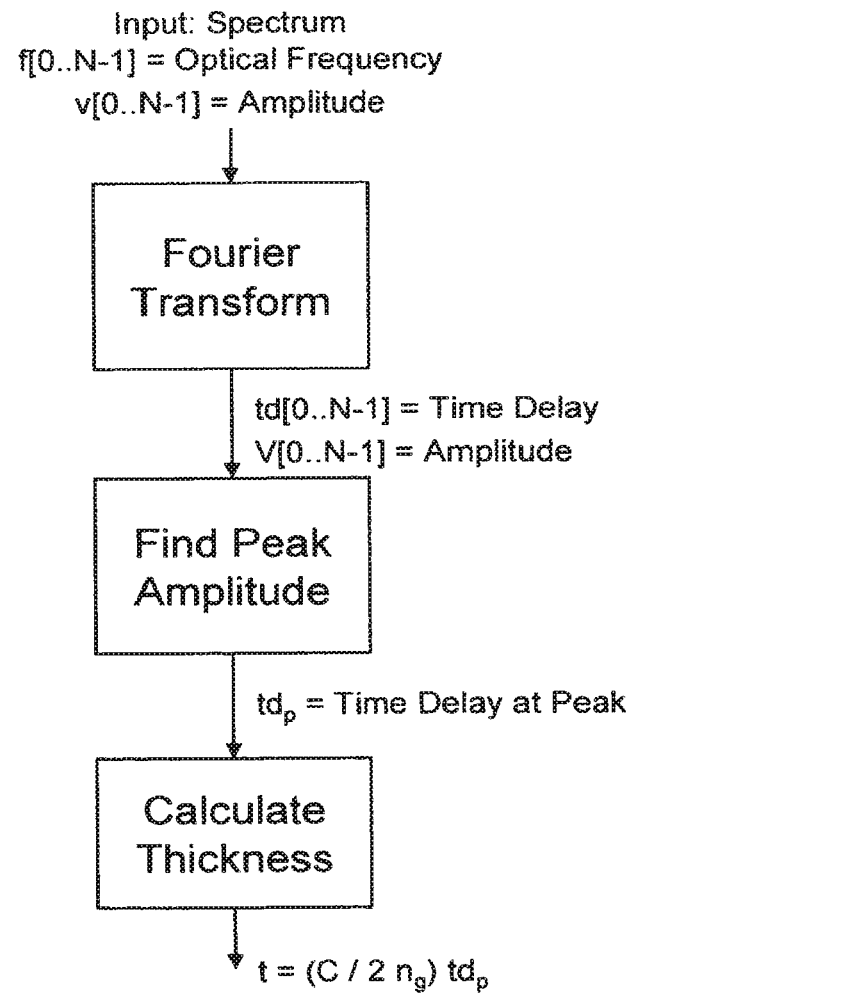
80
86
82
SYSTEM 10

Fig. 3



C = velocity of light in a vacuum (constant)
n_g = group index of refraction of the layer being measured

Fig. 4

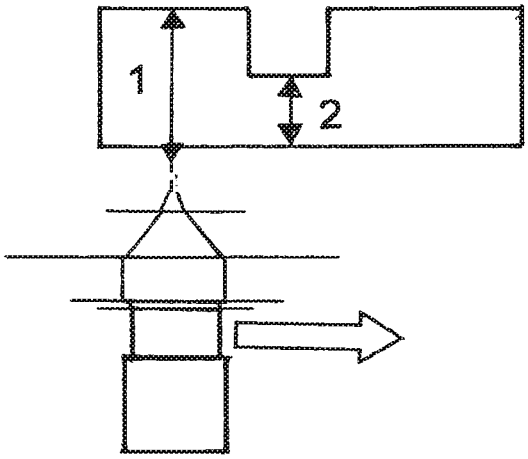


Fig. 5

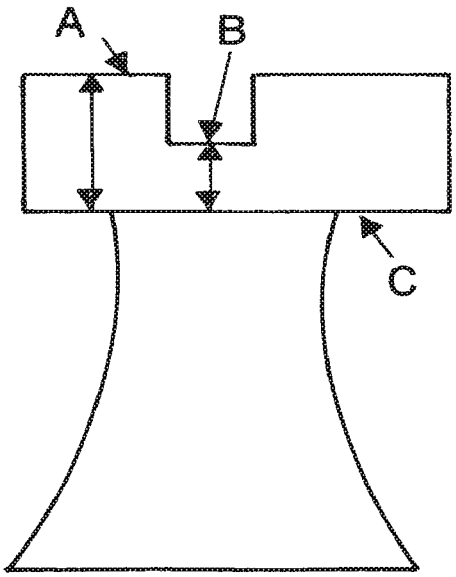


Fig. 6

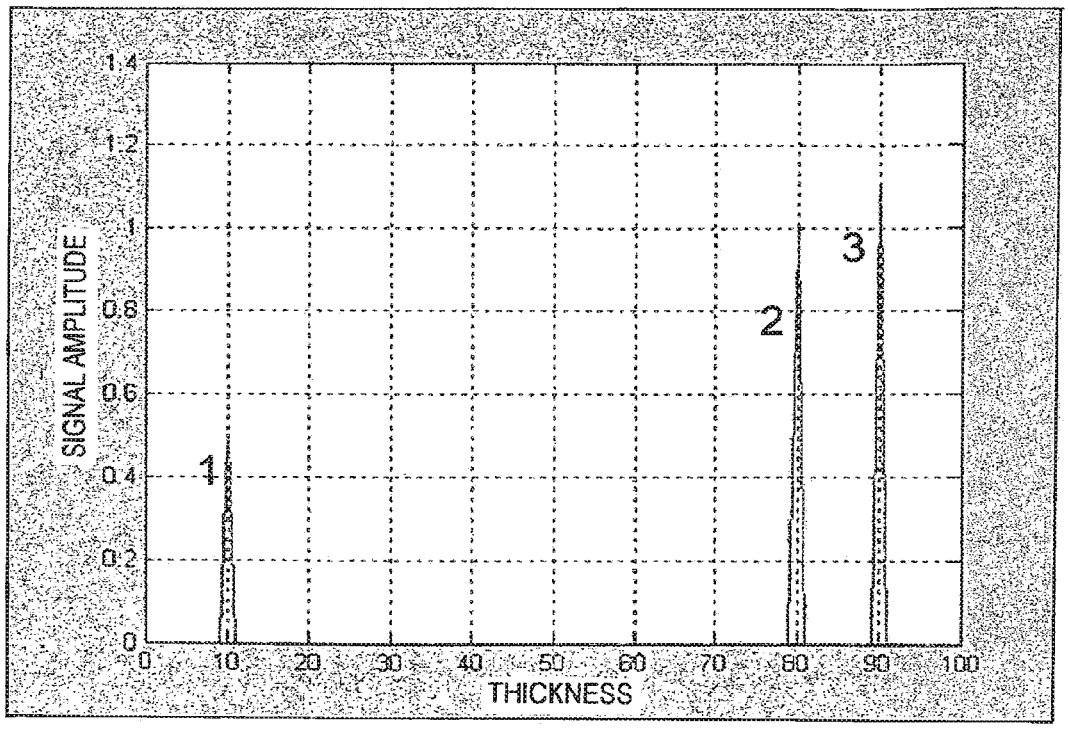

SIGNAL AMPLITUDE
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
10
20
30
40
50
60
70
80
90
100
THICKNESS
1
2
3

Fig. 7

SYSTEM FOR DIRECTLY MEASURING THE DEPTH OF A HIGH ASPECT RATIO ETCHED FEATURE ON A WAFER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. application Ser. No. 14/145,497, filed Dec. 31, 2013, which is a Continuation of U.S. application Ser. No. 13/134,716, filed Jun. 15, 2011, now U.S. Pat. No. 8,649,016, which is a Continuation-in-Part of (C.I.P) of U.S. application Ser. No. 12/456,781 filed Jun. 23, 2009, now abandoned; the contents of all of which are incorporated herein by reference.

TECHNICAL FIELD

The invention generally pertains to systems for measuring an etched feature on a wafer, and more particularly to a system that directly measures the depth of a high aspect ratio etched feature from the non-etched surface of a wafer.

BACKGROUND ART

In the field of electronics, one of the most sought after goals is miniaturization. Efforts to make electronics and electronic memory smaller continue to drive manufactures. There are a variety of techniques and technologies that facilitate continued miniaturization. One technique that is growing in importance is the deep etching of wafers, and the volume of wafers that go through this process is increasing. As the lateral dimensions of the etched features on a wafer, including etched features, continue to decrease, the ability to control the etching process is becoming more difficult.

For companies that make wafers with narrow deep etches, if the aspect ratio, which is defined as the ratio of etched depth to etched width, is more than approximately 5:1, optical non-contact measurement techniques are typically not viable. The reason for this is because optical technologies are not able to receive information from the bottom of an etched feature when viewed from the top. For example, if the width of the etched feature (also referred to as a trench or a via) is smaller than the illumination spot size, then the etched opening itself restricts the amount of light that can illuminate the bottom. However, if the illuminating spot is made smaller by increasing the numerical aperture of the illuminating optics, then the angular spectrum of the illumination will contain light rays at angles to steep to illuminate the bottom of the etched feature. As a result, when the aspect ratio is large and the etched width is small, there is no way to direct a significant portion of the illuminating light on the bottom of the etched feature. Therefore, an engineer who is utilizing the process must take measurements from a scanning electron microscope. These measurements are typically costly, time consuming and most importantly, destructive.

One of the most important electronic devices is a semiconductor, and one of the primary elements of a semiconductor is a silicon on insulator (SOI) wafer. During the manufacturing of electronic devices on a semiconductor wafer, some wafers go through a process known as thinning. There are a variety of reasons why a wafer must be thinned, but the most common are: to dissipate heat quickly, to expose interconnect layers, and to produce the silicon on insulator (SOI) wafer. SOI wafers are comprised of a thin layer of silicon on top of a thin layer of insulator, and these layers are supported by a full thick silicon wafer. Currently, approximately 5% of all semiconductor wafers manufactured are SOI wafers, but this percentage will continue to increase in the future.

The most widely used methods for thinning a wafer are back grinding and polishing. These methods operate on the lower surface, i.e. the non-etched surface, of the wafer. Wafer manufacturers must grind or polish the lower surface of the wafer to the desired thickness with high uniformity. If the wafer is lapped too thin, it might lack the required structural integrity or the interconnects will be overexposed. Even so, the industry trend is directed toward the manufacturing of consistently thinner wafers.

Another important aspect of semiconductor manufacturing is the measurement of wafer thickness. Present technology for measuring wafer thickness includes capacitive sensors, which measure the capacitance of a wafer, and height sensor techniques that utilize two sensors, with one sensor above the wafer and one sensor below the wafer. The use of capacitive sensors typically requires an in-depth knowledge of the wafer material and can generally only function correctly with a single wafer material, not wafers composed of multiple layers of different materials, such as SOI wafers, wafers supported by carriers or handle wafers. Additionally, capacitive sensors have a lower thickness limit of approximately 200 microns and are limited to a small sample of materials.

Height sensor techniques can generally function on thinner samples, and can accommodate multiple layer wafer stacks that are made of virtually any material as they detect the physical surface of a wafer. Height sensor techniques do require delicate alignment in all three axes and require calibration to "teach" the sensor how far apart the axes are in space. This calibration requirement is the lower accuracy limit of the measurements taken.

Present and developing technology for the connection of electronic chips of various functions includes a process known as Through Silicon Vias (TSVs). A TSV is essentially a small cavity in a silicon wafer, with a typical diameter of 1 to 100 microns and a typical depth of 25 to 500 microns. These etched features, as well as others such as trenches, are known as High Aspect Ratio features because their depth is significantly greater than their width. The TSVs are typically filled with a conductive material and then the electronic chips are built on top of the TSVs. When the chips are complete, the wafer is polished, or thinned, from the lower surface until the TSVs are exposed. The TSVs then facilitate the connections that are made to other chips or devices.

During the wafer manufacturing process, the depth of the TSVs is a critical element. If the TSVs are not deep enough, the wafer will not maintain the physical integrity that is required to continue the process when the TSVs are exposed. If the TSVs are too deep, they will be exposed before the wafer is thin enough. A manufacturing process engineer has no means of quickly, accurately and non-destructively measuring the thickness of a wafer during the manufacturing process. Currently, a wafer is thinned until visual inspection confirms the TSVs are exposed, without possessing accurate knowledge of the thickness of the wafer.

There are a variety of requirements for measuring the depth of an etched feature on a wafer. The high aspect ratio etched features are difficult to measure when using optical techniques because light can not propagate to the bottom of an etched feature and then return. This applies to both confocal techniques as well as interferometric techniques. Optical techniques are limited to an aspect ratio of 2 or 3 to 1. For etched features with a higher aspect ratio, the only current method to directly measure the depth is by destruc-

tively sectioning the wafer and then viewing the etched feature from the side. This method is undesirable because the wafer sample is destroyed and the electron microscope that must be utilized is expensive and time consuming. Another wafer thinning process occurs for Silicon On Insulator (SOI) applications. SOI is essentially a wafer stack that comprises the fusing of two wafers with a thin oxide layer between the two wafers. The first wafer, which is known as the device wafer is then polished to a specific thickness. The second wafer, which is known as the handle wafer, is sacrificial and is used simply to maintain the structural integrity of the device wafer.

Current technology only measures the entire wafer stack and can not differentiate between the two wafers. Therefore, a process engineer must measure the first/handle wafer, measure the oxide layer and then subtract these measurements in order to calculate the thickness of the second/device layer. The sensor utilized in the instant invention can differentiate between the two wafers and can measure the first/device layer directly, thereby reducing the error budget and improving accuracy.

A search of the prior art did not disclose literature or patents that read directly on the claims of the instant invention. However, the following U.S. patents are considered related.

PAT. NO.	INVENTOR	ISSUED
US2007/0148792(Pub)	Marx, et al	28 Jun. 2007
6,878,301	Mundt	12 Apr. 2005
6,822,745	DeGrout, et al	23 Nov. 2004
6,806,105	Johnson, et al	19 Oct. 2004

The US2007/0148792 Publication discloses a system for measuring wafer thickness, flatness and the depth of etched features thereon. The system utilizes the non-etched surface of the wafer to accurately measure the lower surface of an etched feature, rendering the etched feature an effective bump that can be measured on the etched surface and non-etched surface by using a non-contact optical instrument. The system discloses the use of distance measuring sensors such as a chromatic confocal sensor to measure the trench, as a bump. The system utilizes a light source that renders the material of which the wafer is composed transparent. The light source functions in the near infrared region for measuring the thickness and trench depth of wafers made of silicon, which is opaque in the visible region and transparent in the rear infrared region. The difference between the referenced publication and the instant application is that the referenced publication uses distance measuring sensors, whereas the instant application uses a reflectometer.

The U.S. Pat. No. 6,878,301 patent discloses a method and an apparatus for optically detecting a trench's depth by detecting a first maxima in an intensity of multi-wavelength light. A portion of the multi-wavelength light is reflected from the trench's top surface. A second maxima in an intensity of multi-wavelength light is also detected. A portion of the multi-wavelength light is reflected from the trench's bottom surface. A maxima peak separation between the first maxima and the second maxima is determined which corresponds to the maxima peak separation. The apparatus is essentially a Michelson interferometer, and as such, requires a reference mirror. Additionally, the method disclosed illuminates the wafer on the etched surface.

The U.S. Pat. No. 6,822,745 patent discloses a method for determining a geometric property of a test object. The method includes interferometrically profiling a first surface of the test object in a first coordinate system; interferometrically profiling a second surface of the test object in a second coordinate system that is different from the first coordinate system; and providing a relationship between the first and second coordinate system. The geometric property is calculated based on the interferometrically profiled surfaces and the relationship between the first and second coordinate system.

The U.S. Pat. No. 6,806,105 patent discloses a method for measuring a parameter associated with a portion of a sample. The sample has formed thereon one or more structures with at least two zones each having an associated zone reflectance property. The method includes illuminating the zone; with broadband light, and measuring at least one reflectance property of light reflected from the two zones. The method includes fitting a parametric model to the measured reflectance property. The parametric model mixes the reflectance properties of the zones to account for partially coherent light interactions between the two zones.

Publication

C. A. Duran, et al., "Infrared Reflectometery for Metrology of Trenches in Power Devices," 2007 IEEE/SEMI Advanced Semiconductor Manufacturing Conference p. 175.

Duran discloses a method of measuring the characteristics, including depth, of etched trenches in silicon. The method uses an illumination spot many times larger than the width of the etched trenches. The method illuminates the trenches using infrared light with wavelengths in the range from 1.4 microns to 20 microns. The illumination is on the etched side of the wafer. The reflected intensity is measured at different wavelengths and angles and compared to predicted measurements calculated with the rigorous coupled wave analysis (RCWA) method. The measurement results are the trench parameters used in the calculation that produce predicted measurements most similar to the actual measurements. Because the method relies on RCWA, the method is only valid for periodic structures of trenches, since it only measures the average characteristics for all of the trenches; it cannot measure the characteristics of an individual trench, nor of a single isolated trench. Furthermore, the publication describes trenches with an aspect ratio of approximately 2:1, and since the illumination is on the side of the etched surface, it will fail to illuminate the bottom of a trench with higher aspect ratio. Additionally, the method uses an illumination spot many times larger than the width of the etched trenches.

Overview

A system is disclosed for measuring the depth of a high aspect ratio etched feature, such as a trench, which is located on a wafer. In its basic design configuration, the system uses an infrared reflectometer with a small spot size that is aligned to illuminate an etched feature from the non-etched surface of the wafer.

While the principles of reflectometry, and the use of reflectometers for measuring the thickness of layers are well known, the system utilizes a reflectometer for the direct measurement of the depth of a high aspect ratio etched feature, as well as the thickness of multiple layers of a wafer. Unlike Model-Based Infrared Reflectometry (MBIR), the system is a direct measurement of an individual etched feature, and unlike standard reflectometry, the system measures etched features with high spatial resolution.

An important aspect of the system is the measurement of etched features from the surface of the wafer opposite the

etched surface. All of the prior art, including MBIR, illuminate and measures the etched features from the etched surface of the wafer.

Reflectometers measure the thickness of transparent layers of a wafer by measuring the wavelength dependence of the Farbry-Perot interference that occurs in the reflected or transmitted light. When a layer is illuminated with a spatially coherent beam of light, both the upper and lower surfaces cause a reflection. These reflections interfere with each other, and the reflected light amplitude can be large or small depending on whether the interference is, constructive or destructive. The interference depends on layer thickness, its index of refraction, and the wavelength of the light: A "layer" is defined as any homogenous medium with approximately parallel upper and lower surfaces. Examples are films deposited on a substrate, and the substrate itself: such as a silicon wafer.

The reflected intensity for a single layer cavity is:

$$R = \left| \frac{r_1 + r_2 e^{i2\delta}}{1 + r_1 r_2 e^{i2\delta}} \right|^2,$$

where r_1 is the reflection coefficient at the first surface of the cavity, r_2 is the reflection coefficient at the second surface, $\delta=2\pi\eta\, t/\lambda$, η is the index of refraction, t is the cavity thickness, and λ is the wavelength. Because of the complex exponential, the reflected intensity is periodic in δ. Since δ is proportional to optical frequency (c/λ), the reflected intensity is periodic in optical frequency with period $c/2nl$, where n is the index of refraction, l is the layer thickness, and c is the speed of light in a vacuum. The reflectometer incorporates a source that consists of many wavelengths or sweeps through many wavelengths in time. The reflected light must be analyzed for its dependence on wavelength. By determining the period of the reflected intensity with optical frequency, (optical frequency is inversely proportional to wavelength), the thickness of a wafer layer can be determined, if the index of refraction is also known. Reflectometers can be used to measure the thickness and index of a transparent layer. If the light source uses infrared wavelengths, and specifically wavelengths longer than 1.2 μm then the thickness of a silicon wafer can be measured with a reflectometer.

When the cavity consists of multiple layers, the equation for the reflected intensity contains terms that combine the various combinations of thickness. The resulting reflected spectrum contains several periodicities, such as the period corresponding to the optical thickness of each individual layer, and the period corresponding to the total optical thickness.

To produce a sufficient interference signal, which is observed through the fringe visibility of the reflected signal, the reflections from the wafer's upper and lower surface must have similar beam divergences. For example, if the incident beam is collimated, then the reflected beams from the two surfaces will be collimated, and the interference will be strong and fringe visibility high. However, the present invention requires the incident beam to be focused so that only a small area is illuminated and the etched feature can be spatially distinguished. For substrates that are thicker than the depth of focus, the two reflections have different divergence angles, and so the fringe visibility is reduced. The depth of focus of a focused beam in a medium is approximately proportional to:

$$DOF = \frac{\lambda n}{2NA^2},$$

where λ is the wavelength, n is the index of refraction of the medium, and NA is the numerical aperture of the focusing lens. Thus, the depth of focus increases with the index of refraction of the medium, which is approximately 3.6 for silicon. For example, if the numerical aperture is 0.1, and the wavelength 1.31 μm then the depth of focus in silicon is approximately 468 μm. As a result, a silicon wafer can be as thick as 1000 μm and the fringe visibility would still be strong enough to provide a good signal.

The diameter of the focused spot relative to the diameter or width of the etched feature affects the quality and signal-to-noise ratio of the etch depth measurement. The diameter of the focused spot is approximately proportional to:

$$D=\lambda NA$$

and is independent of the index of refraction of the substrate. If the measurement spot is smaller than the diameter or width of the etched feature, then the whole of the spot reflects off the bottom of the feature and contributes to the thickness measurement signal. When the diameter or width of the feature is similar to or less the measurement spot diameter, only a portion of the incident light contributes to the thickness measurement, and so the signal quality is reduced. At some feature dimension, the reflected signal will be too weak to be useful.

By placing the sensor so that it faces the non-etched surface, the etched feature appears to be a bump in the etched surface of the wafer from the point of view of the sensor. Thus, there is no sidewall interference that inhibits optical techniques now employed and the bottom of the etched feature is easily illuminated regardless of the etched feature's depth.

When the wafer sample consists of multiple layers, the Fabry-Perot interference causes the reflected signal to consist of multiple periodicities in optical frequency. This signal can be analyzed to recover the wafer's material thickness as well as the thicknesses of the other layers. If an intermediate layer also includes an etched feature that would normally be hidden from view by a second layer, the instant invention measures the dimensions of the etched feature by measuring the variation in thickness of the etched layer. If the etched trench has a diameter smaller than the diameter of the illumination spot, then the layer from the etched surface of the wafer to the bottom of the etched trench behaves as a separate layer in the reflectometer signal. Therefore, in this circumstance, the reflectometer simultaneously measures the full thickness of the wafer, the thickness from the bottom of the trench to the non-etched surface, and from the etched surface to the bottom of the etched trench (etched depth).

In view of the above disclosure the primary object of the system is to provide a system that directly measures from the non-etched surface of a wafer, the depth of a high aspect ratio etched feature, such as a trench, which is located on a wafer.

In view of the above disclosure the primary object of the system is to provide a system that directly measures from the non-etched surface of a wafer, the depth of a high aspect ratio etched feature, such as a trench, which is located on a wafer.

In addition to the primary object of the system it is also an object of the system to provide a system that:

can be used for wafers having various dimensions,

is transportable,

is relatively easy to learn and operate when compared to similar systems,

is fast, reliable and relatively maintenance free,

can be custom configured for a user's specific needs,

is accurate, repeatable, non-destructive and has a high spatial resolution, and

is cost effective from both a manufacturer's and consumer's point of view.

These and other objects and advantages of the present invention will become apparent from the subsequent detailed description of the preferred embodiment and the appended claims taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram showing the first embodiment that depicts a basic design of a system for directly measuring the depth of a high aspect ratio etched feature on a wafer.

FIG. **2** is a block diagram showing the second embodiment of the system.

FIG. **3** is a block diagram showing the third embodiment of the system.

FIG. **4** is a flow diagram of the system software.

FIG. **5** is a block diagram showing a sequential scanning method.

FIG. **6** is a block design showing a direct scanning method.

FIG. **7** is a graph depicting the thickness and the signal amplitude for a method wherein the width of the etched feature is smaller than the illumination spot size.

DETAILED DESCRIPTION

System **10** directly measures the depth of a high aspect ratio etched feature located on a wafer is presented in terms of a first basic embodiment, a second embodiment and a third embodiment, as shown respectively in FIGS. **1-3**. The computer software **32** that operates the system **10** is illustrated as a flow diagram in FIG. **4**. The wafer **80**, as also shown in FIGS. **1-3**, includes an etched surface **82** and a non-etched surface **84**, wherein the etched surface **82** is also referred to as a trench **86**.

The first basic embodiment of the system **10**, as shown in FIG. **1**, is comprised of the following major elements: a reflectometer **12**, an analog-to-digital converter (ADC) **24**, a computer **30**, and software **32**. The methods for operating the system **10** are described and shown in FIGS. **5**, **6** and **7**.

The reflectometer **12**, which is preferably comprised of an infrared reflectometer **12**, is positioned to face the non-etched surface **84** of the wafer **80**. The reflectometer **12** has means for producing a focused incident light **23** that is applied to the non-etched surface **84** of the wafer **80**. The reflectometer **12** also has means for receiving and processing a light **25** that is reflected from the non-etched surface **84** of the wafer **80**, as shown in FIG. **1**. The output of the reflectometer **12** is an analog optical frequency spectrum signal **27** that is applied to the ADC **24**, where the analog optical frequency spectrum signal **27** is converted to a corresponding digital data signal **29**.

The computer **30**, in combination with the software **32**, has means for processing the digital data signal **29** applied from the ADC **24** and to display the depth of the high aspect ratio etched surface of the wafer **80**. The computer **30** can also produce a display signal **31** that is applied to an optional display **34** (not shown) where the depth of the high aspect ratio etched feature of the wafer **80** can be displayed.

The infrared reflectometer **12** in a preferred configuration is comprised of a swept laser **14**, a fiber circulator **16** or a beam splitter **26** (not shown), a collimator **18**, an objective lens **20**, and a photodetector **22**. The swept laser **14**, which sweeps through range of wavelengths in time, is connected to the fiber circulator **16** or to the beam splitter **26**, either of which is connected to both the collimator **18** and the photodetector **22**. The collimator **18** functions in combination with the objective lens **20** which is positioned above the non-etched surface **84** of the wafer **80**.

The broadband light source can be comprised of either the swept laser **14**, as shown in FIGS. **1** and **2**, or a broadband incoherent infrared source **40**, as shown in FIG. **3**. The broadband incoherent infrared source **40** can be selected from the group consisting of an LED, a halogen light source, a mercury arc lamp or a superluminescent diode (SLED). When the photodetector **22** is used, the light source is comprised of the swept laser **14**, as shown in FIG. **2**; when a spectrometer **42** is used, the light source is comprised of broadband incoherent infrared source **40**, as shown in FIG. **3**. In all cases, the broadband light source produces a set of wavelengths that encompass a range wherein the wafer **80** is transparent to the wavelengths.

The software **32** is applied the digital data signal **29** from the ADC **24** and performs the following operations:

a) a Fourier transform,

b) find peak amplitude, and

c) calculate thickness.

As in the second and third embodiments, the first embodiment can utilize a wafer holding fixture **36**. The fixture **36** has means for being laterally displaced and means for positioning the non-etched surface **84** of the wafer **80** facing the infrared reflectometer **12**.

The second embodiment which is the preferred embodiment of the system **10**, as shown in FIG. **2**, is comprised of the following elements: an infrared reflectometer **12**, a swept laser **14**, a fiber circulator **16**, a collimator **18**, an objective lens **20**, a photodetector **22**, an ADC **24**, a computer **30**, software **32**, a display **34** and a wafer holding fixture **36**.

The infrared reflectometer **12** is designed to produce a focused incident light **23**, to receive a reflected light **25**, and to indicate their ratio. The reflectometer **12** uses a small spot size to illuminate the non-etched surface **84** of the wafer **80**. The reflectometer **12** in the second embodiment is comprised of the swept laser **14** which produces a set of wavelengths **11** that are swept in time, a trigger signal **13** and a clock signal **15**. The swept wavelengths **11** typically range from 1260 mm to 1360 mm which encompasses a range where the wafer **80** is transparent.

The fiber circulator **16** has a first input that receives the range of wavelengths **11** from the swept laser **14**. The fiber circulator has a first light output **19** which directs the light to the collimator **18**. The collimator **18** collimates the light which is then focused by the objective lens **20** onto the non-etched surface **84** of the wafer **80**. The reflected light **25** from the wafer **80** is composed of reflections from the non-etched surface **84**, the etched surface **82**, and any other layers that comprise the wafer. The reflected light **25** is collected by the objective lens **20** and returned to the fiber circulator **16** via the collimator **18**. The reflected light return **21** applied to the fiber circulator **16** and the first light output **19** from the fiber circulator **16** share the same physical optical fiber. The reflected light **25** is then directed by the second light output **17** from the fiber circulator **16** to the photodetector **22**. The photodetector **22**, when applied the

second light output **17** from the fiber circulator **16**, produces an analog optical frequency spectrum signal **27** that is applied the ADC **24**.

The ADC **24**, which is designed to function as a data acquisition device, has a first input and a second input, that are applied respectively from the swept laser **14**, the trigger signal **13** and the clock signal **15**, as shown in FIG. **2**. The ADC **24** further has a third input that is applied the analog optical frequency spectrum **27** from the photodetector **22**, wherein the output of the ADC **24** is a corresponding digital data signal **29**.

The computer **30**, in combination with the software **32**, has means for processing the digital data signal **29** applied from the ADC **24** and to produce a display signal **31** that portrays on the display **34** the high aspect ratio etched feature of the wafer **80**.

The wafer holding fixture **36**, as shown in FIG. **3**, has an open area **38** that is larger than the etched surface **82** of the wafer **80**, a means for being laterally displaced, and a means for placing the non-etched surface **84** of the wafer **80** facing the objective lens **20** located on the infrared reflectometer **12**. The wafer **80** is selected from the group consisting of Si, GaAs, GaAlAs, InP, SiC, SiO_2 and sapphire.

The third embodiment, as shown in FIG. **3**, is comprised of the following elements: a broadband incoherent infrared source (BITS) **40**, a fiber circulator **16**, a collimator **18**, an objective lens **20**, a spectrometer **42**, a computer **30**, software **32**, a display **34** and a wafer holding fixture **36**.

The broadband incoherent infrared source (BITS) **40** has means for producing an incoherent light **35** that is applied to the first input of the fiber circulator **16**. The fiber circulator has a first light output **19** which is directed the light to the collimator **18**. The collimator **18** collimates the light which is then focused by the objective lens **20** to produce a focused incident light **23** that is applied to the non-etched surface **84** of the wafer **80**. From the wafer **80** is produced a reflected light **25** that is composed of reflections from the non-etched surface **84**, the etched surface **82**, and any other layers that might comprise the wafer **80**. The reflected light **25** is collected by the objective lens **20** and returned to the fiber circulator **16** by means of the light return **21** and the first light output **19** from the fiber circulator **16** share the same physical optical fiber. A light signal **37** is directed from the fiber circulator **16** to the spectrometer **42**, as shown in FIG. **3**.

The spectrometer **42**, which can consist of a grating spectrometer, a Fourier transform infrared spectrometer (FTIR), or other types of spectrometers, is applied the light signal **37** from the fiber circulator **16**. From the spectrometer **42** is produced a light spectrum signal **39** that is applied to the computer **30**. The computer **30**, in combination with the software **32**, has means for processing the light spectrum signal **39** and to produce a digital data signal **29**. The digital data signal **29** portrays on the display **34** the depth of the high aspect ratio etched feature of the wafer **80**.

The wafer holding fixture **36**, as shown in FIG. **3**, has means for being laterally displaced and means for positioning the non-etched surface **84** of the wafer **80** facing the objective lens **20**.

When the width of the etched feature is equal to or greater than the illumination spot size, a sequential scanning method is preferred. To measure the depth of an etched feature with this method, first, measure the thickness of an adjacent area of the wafer **80** that is not etched, as designated by the numeral "**1**" in FIG. **5**; second, measure the thickness of the wafer **80** where it is etched (thickness from the bottom of the etched feature to the non-etched surface **84** of the wafer **80**), as designated by the numeral "**2**" in FIG. **5**; and third, subtract the two measurements.

When the width of the etched feature is smaller than the spot size, e.g., the etched feature is a circular hole with a 5 μm diameter and the illumination spot size is 13 μm, then the etched depth can be measured with a direct method. Since the illumination covers the etched feature, as designated by the letter "B" in FIG. **6**, and the surrounding wafer surface, as designated by the letter "A" in FIG. **6**, light is reflected from both of the surfaces A and B. The reflected light from A and B interferes with light reflected from the non-etched surface of the wafer, as designated by the letter "C" in FIG. **6**. An example of the measurement signals is shown in FIG. **7**. In this example, Peak #**1** in FIG. **7** results from the interference of the light reflected from the bottom of the etched feature, as designated by the letter "B" in FIG. **6**, and the top of the etched surface of the wafer **80**, as designated "A" in FIG. **6**, and so Peak #**1** is a direct measurement of the etched depth. Therefore, in one measurement instant, without requiring mechanical scanning or the manipulation of separate measurements, a direct measurement of the etched depth is possible.

The thickness measurement for the method where the width of the etched feature is smaller than the illumination spot size is shown in FIG. **7**. Peak **1** corresponds to the interference between the reflections from surfaces A and B of FIG. **6**; Peak **2** corresponds to the interference between the reflections from surfaces B and C of FIG. **6**; and Peak **3** measures the full wafer thickness from the interference between the reflections from surfaces A and C, as also shown in FIG. **6**.

The first method for measuring the depth of a high aspect ratio etched feature on a wafer **80** having an etched surface **82** and a non-etched surface **84** is typically utilized when the width of the etched feature **86** is equal to or greater than the applied illumination spot size. The first method, which is also referred to as a sequential scanning method, is conducted by performing the following steps:

a) position a reflectometer facing the non-etched surface of the wafer,

b) measure the thickness of the wafer **80** at a location where it is not etched,

c) measure the thickness of the etched surface **82** of the wafer **80** at a location where it is etched, and

d) subtract the two measurements to determine the depth of the etched feature.

The second method for measuring the depth of a high aspect ratio etched feature on a wafer **80** having an etched surface **82** and a non-etched surface **84** is utilized when the width of the etched surface **82** is smaller than the diameter of the applied illuminating spot size. The second method, which is also referred to as the direct method, is conducted by performing the following steps:

a) position a reflectometer facing the non-etched surface of the wafer,

b) simultaneously illuminate the etched feature **86**, the etched surface **82** adjacent to the etched feature **86**, and the non-etched surface **84**, and

c) directly measure the interference between the reflections from the etched surface **82** and the etched feature **86**, and

d) calculate the etch depth directly from the interference recorded in step c).

While the invention has been described in detail and pictorially shown in the accompanying drawings it is not to be limited to such details, since many changes and modifications may be made to the invention without departing

from the spirit and the scope thereof. Hence, it is described to cover any and all modifications and forms which may come within the language and scope of the claims.

The invention claimed is:

1. A system for directly measuring a depth of a high aspect ratio etched feature on a semiconductor wafer having an etched surface opposite a non-etched surface, the high aspect ratio etched feature being etched into the etched surface, the system comprising:

a fixture for supporting the semiconductor wafer and for moving the semiconductor wafer in an XY plane;

a sensor having an optical axis that is perpendicular to the XY plane in which the fixture moves the semiconductor wafer, the sensor further comprising:

an illumination source that directs light onto the non-etched surface of the semiconductor wafer, wherein the directed light is incident upon both the non-etched surface and the etched feature of the semiconductor wafer;

a light capturing device arranged to capture light that is returned from the semiconductor wafer;

a photodetector configured to convert the captured light into an optical frequency spectrum; and

a computing device programmed to:

determine a first thickness of the semiconductor wafer at a region where the semiconductor wafer is not etched based upon an analysis of information indicative of the optical frequency spectrum,

determine a second thickness of the semiconductor wafer at a region of the etched feature based upon an analysis of information of the optical frequency spectrum,

determine the depth of the etched feature by subtracting the second thickness from the first thickness.

2. The system of claim **1**, wherein the illumination source is configured to focus the directed light to define an illumination spot on the non-etched surface, and further wherein a size of the illumination spot is not greater than a size of the etched feature.

3. The system of claim **2**, wherein a diameter of the illumination spot is not greater than a width of the etched feature.

4. The system of claim **2**, wherein a diameter of the illumination spot is not greater than a diameter of the etched feature.

5. The system of claim **2**, wherein the illumination source is configured to sequentially locate the illumination spot at the region where the semiconductor wafer is not etched followed by locating the illumination spot at the region of the etched feature.

6. The system of claim **5**, wherein the computing device is further programmed to determine the first thickness of the semiconductor wafer at the region where the semiconductor wafer is not etched based upon an analysis of information indicative of the optical frequency spectrum of captured light returned from the semiconductor wafer when the illumination spot is located at the region where the semiconductor wafer is not etched.

7. The system of claim **6**, wherein the computing device is further programmed to determine the second thickness of the semiconductor wafer at the region of the etched feature based upon an analysis of information indicative of the optical frequency spectrum of captured light returned from the semiconductor wafer when the illumination spot is located at the region of the etched feature.

8. The system of claim **1**, wherein the computing device is further programmed to determine the first thickness of the semiconductor wafer at the region where the wafer is not etched further based upon an index of refraction of the semiconductor wafer.

9. The system of claim **8**, wherein the computing device is further programmed to determine the second thickness of the semiconductor wafer at the region of the etched feature further based upon the index of refraction of the wafer.

10. The system of claim **1**, wherein the photodetector device is configured to generate the optical frequency spectrum in analog form, and further wherein the system further comprises an analog-to-digital converter (ADC) electronically connected to the photodetector and programmed to generate the optical frequency spectrum in digital form, and even further wherein the computing device is further programmed to determine the first thickness of the semiconductor wafer at the region where the semiconductor wafer is not etched by analyzing the optical frequency spectrum in digital form.

11. The system of claim **10**, wherein the computing device is further programmed to determine the second thickness of the semiconductor wafer at the region of the etched feature by analyzing the optical frequency spectrum in digital form.

12. The system of claim **1**, wherein the light capturing device includes a reflectometer arranged to face the non-etched surface of the semiconductor wafer, wherein the reflectometer is configured to direct light from the illumination source onto the non-etched surface and capture light returned from the wafer.

13. The system of claim **1**, wherein the computing device is further programmed to determine the first thickness of the semiconductor wafer at the region where the semiconductor wafer is not etched by determining a full thickness of the semiconductor wafer between the etched and non-etched surfaces.

14. The system of claim **1**, wherein the etched feature extends from the etched surface to a bottom opposite the etched surface, and further wherein the computing device is further programmed to determine the second thickness of the semiconductor wafer at the region of the etched feature by determining a thickness of the semiconductor wafer between the bottom and the non-etched surface.

15. The system of claim **1**, wherein the etched feature is a trench.

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